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APPLIED AI & GOVERNED ANALYTICS · CASE STUDY

BI Operations Agent: Safe NL-to-SQL & KPI Monitor

BI Scientist • AI/ML Engineer • Python Developer

Natural-language analytics agent with a governed YAML semantic layer, AST read-only SQL guard, 96% in-sample accuracy (56% on held-out questions), and rolling MAD anomaly monitoring.

THE TEXT-TO-SQL PROBLEM

Why Raw LLM Text-to-SQL Destroys Business Truth

Standard text-to-SQL tools plug raw database schemas into LLM prompts, leading to catastrophic hallucinations and security breaches.

Metric Hallucination

- LLMs invent contradictory formulas for `gross_margin` across questions.
- Joins across complex star schemas produce inflated cartesian duplicates.
- Business leadership gets different numbers for the exact same metric.

SQL Injection Risks

- Adversarial prompts trick LLMs into generating `DROP TABLE` or `UPDATE`.
- Multi-statement `;` attacks bypass naive regex checks.
- Sensitive PII is exposed to unauthorized users.

High Latency & Costs

- Giant prompt schemas waste thousands of tokens per query.
- Unindexed multi-table queries cause database lockups.
- Average query latency exceeds 5 to 10 seconds.

GOVERNED ARCHITECTURE

Semantic Layer YAML + AST Read-Only SQL Guard

Separating business metric definitions from query execution for 100% mathematical consistency.

YAML Semantic Layer

- Governed metrics: revenue, gross margin %, return rate.
- Approved dimensional hierarchies & tables.
- Deterministic pattern compiler fallback.

AST SQL Safety Guard

- Single statement starting with SELECT / WITH.
- Blocks all DDL, DML, and table traversals.
- Enforces table allow-lists and LIMIT 500.

MAD Anomaly Monitor

- Rolling 28-day robust MAD z-scores.
- Flags store & category KPI deviations ($|z| \geq 2.5$).
- Compiles weekly executive markdown digest.

OPERATIONS INTERFACE

BI Agent Interactive Dashboard

FastAPI service and Streamlit application querying 36,550 retail transactions.

- **Plain-English Copilot:** Business users ask questions in natural language and receive verified SQL, tables, and Vega-Lite chart specs.
- **Automated KPI Digest:** Generates weekly executive summaries highlighting top-performing stores and anomaly warnings.
- **Fast Embedded Engine:** Queries run locally on SQLite: 17.8 ms median, 36.6 ms p95.

FastAPI | SQLite | Semantic Layer | Streamlit



github.com/ikebude/AIMLOps-BI-Operations-Agent

BENCHMARK VERIFICATION

50-Question Benchmark & Adversarial Security

Evaluated on a 2-year Nigerian multi-store retail dataset (₦10.15B GMV across 36,550 records).

Benchmark Accuracy	Adversarial Injections	Median Latency	p95 Latency
96.0%	**12 / 12 Blocked**	**17.8 ms**	**36.6 ms**
48/50 in-sample; 14/25 held-out	0 unsafe SQL executed	Question to answer, median	95th percentile

Security Verification

- 12 adversarial injection attacks (`DROP TABLE`, `; UPDATE`, `-- comments`) tested in CI.
- Every single attack was intercepted and blocked before execution.
- 20/20 Pytest unit, guard, and evaluation tests passing.

Deterministic Governance

- Metrics are mathematically identical across all users and dashboards.
- Eliminates prompt drift and LLM hallucination in executive reporting.
- Runs on SQLite today; the sqlglot-based guard parses SQL by dialect rather than matching text.

OPEN-SOURCE & GOVERNED

Natural language analytics with mathematical guarantees.

BI Operations Agent delivers the speed of AI with the rigor of enterprise data governance.

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